Global Assembly: Bounded Reduction and Kohler–Jobin Transfer to Sharp Faber–Krahn (Route δ, Plan 2, Building Block GlobalAssembly)

Abstract

We assemble the final three-stage chain that converts the bounded-class sharp Saint–Venant inequality of the Plan 2 metric route into the unconditional sharp quantitative Faber–Krahn inequality. The chain has three stages: (1) the bounded sharp Saint–Venant inequality $\mathcal{A}(\Omega)^2 \le C(n,R,\rho_*)\,\delta_T(\Omega)$ supplied by [6]; (2) the bounded-to-global reduction of [7, Lemmas 5.1–5.3], which internalises the R-dependence at the dimensional radius $R_*(n) := \max\{d(n),2\}$ and emits a dimensional Saint–Venant constant $c_{\mathrm{SV}}^{\mathrm{glob}}(n) > 0$; and (3) the Kohler–Jobin transfer of [8], which converts sharp Saint–Venant into sharp Faber–Krahn with constant $c_{\mathrm{FK}}(n) > 0$. Stages 2 and 3 are cited verbatim from the Plan 1 Final/ library; the distinctive feature of Plan 2 is that the bounded input to stage 2 is delivered by the metric level-set route (no selection principle, no graph entry, no Schauder interpolation, no nearly-spherical theorem).

1 Statement of the main theorem

Throughout $n \ge 2$. We use the standard normalisations: $u_\Omega \in W_0^{1,2}(\Omega)$ with $-\Delta u_\Omega = 1$; $T(\Omega) = \int_\Omega u_\Omega$; $E(\Omega) = -T(\Omega)/2$; $\omega_n = |B_1|$, $L_n = \lambda_1(B_1)$, $T_n = T(B_1) = \omega_n/(n(n+2))$; $\mathcal{A}(\Omega) = \inf_{x_0} |\Omega \Delta (x_0 + B^*)|/|B^*| \in [0,2)$; B^* the ball with $|B^*| = |\Omega|$; $\delta_T(\Omega) := E(\Omega) - E(B^*) \ge 0$.

Theorem 1.1 (Sharp quantitative Faber–Krahn, Plan 2)**.** *There exists a dimensional constant $c_{\mathrm{FK}}(n) > 0$ such that for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$\Big(\frac{|\Omega|}{|B_1|}\Big)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \ge c_{\mathrm{FK}}(n)\,\mathcal{A}(\Omega)^2. \tag{1}$$

The exponent 2 is sharp.

2 Stage 1: bounded sharp Saint–Venant

Theorem 2.1 ([6])**.** *For every $n \ge 2$, $R \ge 2$, $\rho_* \in (0,1)$ there exists $C(n,R,\rho_*) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$,*

$$\mathcal{A}(\Omega)^2 \le C(n,R,\rho_*)\,\delta_T(\Omega), \tag{2}$$

equivalently $E(\Omega) - E(B_1) \ge c_{\mathrm{SV}}(n,R,\rho_*)\mathcal{A}(\Omega)^2$ with $c_{\mathrm{SV}} = 1/C$.

This is the Plan 2 metric chain output: LevelSetIdentity $\to$ MetricFramework $\to$ CentroidBound $\to$ SpaceTimeIdentity $\to$ WeightedMetricTrace $\to$ BoundaryLayerTransfer.

Remark 2.2 (Internal parameter ρ_*)**.** We fix $\rho_* = 1/2$ once and for all. Then $c_{\mathrm{SV}}(n,R,1/2)$ depends only on (n,R), and we write $c_{\mathrm{SV}}(n,R)$ henceforth.

Remark 2.3 (Distinction from Plan 1)**.** The same bounded inequality is delivered by Plan 1's SaintVenantAssembly [9] via selection principle + graph entry + Schauder + nearly-spherical closure. Plan 2 delivers it via the metric level-set chain without those tools.

3 Stage 2: bounded reduction

Theorem 3.1 ([7, Thm. 5.1])**.** *Suppose* (2) *holds with constant* $c_{\mathrm{SV}}(n, R_*(n))$ *at the dimensional radius* $R_*(n) := \max\{d(n), 2\}$ *where* $d(n)$ *is the BDV Lemma 5.3 diameter constant. Then for every open* $\Omega \subset \mathbb{R}^n$ *with* $|\Omega| = \omega_n$,

$$E(\Omega) - E(B_1) \geq c_{\mathrm{SV}}^{\mathrm{glob}}(n)\, \mathcal{A}(\Omega)^2, \tag{3}$$

with

$$c_{\mathrm{SV}}^{\mathrm{glob}}(n) = \min\{c_{\mathrm{near}}^{\mathrm{glob}}(n), \omega_n^{(n+2)/n}\delta(n)/16\}, \tag{4}$$

and

$$c_{\mathrm{near}}^{\mathrm{glob}}(n) = \frac{\omega_n^{(n+2)/n}}{4C_9(n)(1/\widetilde{c}_{\mathrm{SV}}(n) + C_9(n))}, \quad \widetilde{c}_{\mathrm{SV}}(n) = \omega_n^{-(n+2)/n} c_{\mathrm{SV}}(n, R_*(n)).$$

By scale invariance, (3) extends to general volume:

$$E(\Omega) - E(B^*) \geq c_{\mathrm{SV}}^{\mathrm{glob}}(n) \Big(\frac{|B^*|}{|B_1|}\Big)^{(n+2)/n} \mathcal{A}(\Omega)^2. \tag{5}$$

4 Stage 3: Kohler–Jobin transfer

Theorem 4.1 ([8, Thm. 6.1])**.** *Suppose* (5) *holds. Then*

$$\Big(\frac{|\Omega|}{|B_1|}\Big)^{2/n} \big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \geq c_{\mathrm{FK}}(n)\, \mathcal{A}(\Omega)^2, \tag{6}$$

with

$$c_{\mathrm{FK}}(n) := \min\Big\{\frac{4L_n c_{\mathrm{SV}}^{\mathrm{glob}}(n)}{(n+2)T_n}, \frac{L_n(2^{2/(n+2)} - 1)}{4}\Big\}. \tag{7}$$

5 Main theorem assembled

Proof of Theorem 1.1. Fix $\rho_* = 1/2$. Apply Theorem 2.1 at $R = R_*(n)$, then Theorem 3.1, then Theorem 4.1 with $c_{\mathrm{SV}}(n) = c_{\mathrm{SV}}^{\mathrm{glob}}(n)$. The output is (1) with $c_{\mathrm{FK}}(n) > 0$ given by (7).

Sharpness: smooth volume-preserving ellipsoidal perturbations of B^* saturate the exponent 2 on $\mathcal{A}$. □

6 Constants chain

Stage	Constant	Dependence
1 output	$C_{B^2}(n, R, \rho_*)$	n, R, ρ_*
ρ_* fix	$\rho_* = 1/2$	n, R
1 at R_*	$c_{\mathrm{SV}}(n, R_*(n))$	n
2 BDV constants	$C_9(n), \delta(n), d(n)$	n
2 dim. radius	$R_*(n) = \max\{d(n), 2\}$	n
2 output	$c_{\mathrm{SV}}^{\mathrm{glob}}(n)$	n
3 KJ multiplier	$2L_n/((n+2)T_n)$	n
3 output	$c_{\mathrm{FK}}(n)$	n

Verdict: All R, ρ_*-dependence absorbed at the Stage 1→2 interface. Final $c_{\mathrm{FK}}(n)$ depends on n alone.

7 Normalisation conventions

Convention	Choice
Torsion energy sign	$E(\Omega) = -T(\Omega)/2$
Saint–Venant deficit	$\delta_T(\Omega) = E(\Omega) - E(B^*)$
Torsion deficit	$\delta_{\mathrm{SV}} = T(B^*) - T(\Omega) = 2\delta_T$
Volume normalisation	$\lvert\Omega\rvert = \lvert B^*\rvert$, proofs at $\lvert\Omega\rvert = \omega_n$
Fraenkel asymmetry	$\mathcal{A}(\Omega) = \inf_{x_0} \lvert\Omega\Delta(x_0 + B^*)\rvert/\lvert B^*\rvert$
Scale-invariant deficit	$D(\Omega) = E\lvert\Omega\rvert^{-(n+2)/n} - E(B_1)\omega_n^{-(n+2)/n}$
Volume radius	$\lvert E_\rho\rvert = \omega_n \rho^n$
Internal ρ_*	$\rho_* = 1/2$
Working radius	$R_*(n) = \max\{d(n), 2\}$
Spectral output	$(\lvert\Omega\rvert/\lvert B_1\rvert)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*))$

The factor 4 in $4L_n c_{\mathrm{SV}}^{\mathrm{glob}}/((n+2)T_n)$ absorbs the factor 2 between δ_{SV} and δ_T.

8 Compatibility with Plan 1

Shared. Stages 2 and 3 are Plan 1 Final/ blocks [7, 8] used verbatim.

Different. Plan 2 delivers Stage 1 via the metric level-set chain [1, 2, 3, 4, 5, 6], which uses:

- No selection principle.
- No graph entry.
- No Schauder interpolation.
- No nearly-spherical closure theorem.

This is the methodological contribution of Plan 2.

9 Open issues

Conditional on the six prior building blocks delivering their stated output, the present block closes unconditionally. The smallness regime restriction is absorbed into Stage 2's near-deficit / far-deficit dichotomy.

The seven building blocks of Plan 2:

1. LevelSetIdentity;
2. MetricFramework;
3. CentroidBound;
4. SpaceTimeIdentity;
5. WeightedMetricTrace;
6. BoundaryLayerTransfer;
7. GlobalAssembly (this block).

References

[1] Plan 2 building block LevelSetIdentity.

[2] Plan 2 building block MetricFramework.

[3] Plan 2 building block CentroidBound.

[4] Plan 2 building block SpaceTimeIdentity.

[5] Plan 2 building block WeightedMetricTrace.

[6] Plan 2 building block BoundaryLayerTransfer.

[7] Plan 1 Final/ block BoundedReduction. `Final/BoundedReduction.tex`.

[8] Plan 1 Final/ block KohlerJobinTransfer. `Final/KohlerJobinTransfer.tex`.

[9] Plan 1 Final/ block SaintVenantAssembly.

[10] L. Brasco, G. De Philippis, B. Velichkov, *Faber–Krahn inequalities in sharp quantitative form*, Duke Math. J. **164** (2015), 1777–1831.

[11] L. Brasco, ESAIM Control Optim. Calc. Var. **20** (2014), 315–338.

[12] L. Brasco, G. De Philippis, in *Shape Optimization and Spectral Theory*, De Gruyter Open, 2017.

[13] M. Allen, D. Kriventsov, R. Neumayer, *Sharp quantitative Faber–Krahn inequalities and the Alt–Caffarelli–Friedman monotonicity formula*, Ars Inveniendi Analytica (2023).

[14] N. Fusco, F. Maggi, A. Pratelli, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) **8** (2009), 51–71.